

CMSC 104 Section 2

Fall 2026

Sample Exam for Exam 1

Purpose:

This sample exam is created to give students an understanding of what Exam 1 will look like. The number, type and point values of the questions on this exam are the same as the actual exam.

Students will be given 75 minutes to complete the actual exam. Students can use that as a guide in working on this sample exam to ensure that they will be able to complete the exam in time.

Instructions:

This is an open-book test. "Open book" means that students can use laptops, notes, books, IDEs or other sources to help answer a question. There are two sources students may NOT use: other students, and AI. Students are not permitted to communicate with any other student, whether in the classroom or not. No talking, copying, messaging or other communication is permitted. Students are also prohibited from using AI sources such as Gemini, ChatGPT, Claude, Grok or others to answer questions.

Total: This test is worth 100 points.

Section 1: True/False and Multiple Choice (40 points)

This section has 10 questions, each of which is worth 4 points. NO PARTIAL CREDIT is given for questions in this section. Answers are either correct, or incorrect.

1. Which of the following represents computer software instead of hardware?
 - a. A monitor
 - b. A mouse
 - c. Windows 11
 - d. A keyboard

2. True or False: Microsoft Excel is an example of System software
 - a. True
 - b. False
3. True or False: $3 / 4$ is the same as $3 // 4$
 - a. True
 - b. False
4. What's the best way to develop computer software?
 - a. Write the entire program in a single shot and hope like heck it works
 - b. Write a little, test a little
 - c. The answer is b
 - d. I'm serious, just go ahead and pick b
5. What's the result of the following for i: loop in Python?

```
import math
for j in range(int(math.log2(k))):
    k /= 2
print(k)
```

 - a. Nothing; the loop is invalid and produces an error
 - b. 1.0
 - c. 3.14159
 - d. None of the above
6. Which of the following is true about Python input statements?
 - a. The input statement prints whatever the prompt is on the screen, and then waits for the user to type the requested input. The program does not resume until the user hits the Enter key
 - b. The prompt must be exactly one string (where one integer or one float will be converted to string so they'll work too)
 - c. All input comes into the program as strings, so if you need integers or floats you have to convert
 - d. All of the above are true
7. What's the difference between `print("UMBC"+"Retrievers")` and `print("UMBC","Retrievers")`
 - a. There is no difference; they produce identical output
 - b. The statement with the + prints one blank space between "UMBC" and "Retrievers" while the statement with the , does not
 - c. The statement with the , prints one blank space between "UMBC" and "Retrievers" while the statement with the + does not
 - d. None of the above is true

8. If x is a boolean variable, the statement x = "True" is a valid assignment statement.
 - a. True
 - b. False
9. What is the value of 5**6**7
 - a. 5**(279936)
 - b. 15625**7
 - c. Both of the above are correct
 - d. None of the above is correct
10. True or False: a solid-state drive (SSD) is computer software
 - a. True
 - b. False

Section 2 Short answer (40 points)

This section has 8 questions, each of which is worth 5 points. Partial Credit WILL be awarded for questions in this section. Note that when we say "short" answer we mean that the questions can usually be answered by one or two sentences; three at most. If you write an entire paragraph you're not really helping your case.

11. Give one reason why high-level languages like Python are better for programmers than machine languages?
12. What are the only two values that computers understand internally?
13. What three values are found in each entry in the Python symbol table?
14. Why doesn't 5+3%4 equal zero?
15. The following code:

```
i = 10
for j in range (11,1, -2):
```

```
i += j  
print (i)
```

Produces the value 45. What five values are added to i in the loop to produce this value?

16. Suppose you are asked to write a Python program to calculate the inverse hyperbolic tangent of a quantity. Why should you import the math library to solve this problem instead of just writing the code yourself?

17. What is a boolean variable in Python?

18. Is 5.0E26 a valid integer value in Python?

Section 3 Programming (20 points)

This section has 2 questions, each of which is worth 10 points. Partial Credit WILL be awarded for questions in this section.

19. Write a Python program that asks the user for three values: their last name, their major field, and their GPA. Use three different input statements to get these values. Then, print out each value along with an explanation of what you are printing.

20. Write a program that calculates the value of 10!. Use a “for i” loop in your program. In other words, you can’t just write `print(10*9*8*7*6*5*4*3*2*1)`. There is no user input. Print out the value you calculate, with an explanation for the user of what your program has done.